Monetary Policy Transmission through Cross-Selling Banks*

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Abstract

Banks often expect to later sell loans to their depositors. A prospective borrower is most valuable when policy rates are low, so banks compete harder for high-potential depositors when rates fall. Such depositors receive higher deposit rate levels but lower pass-through. In a model and in administrative data on every Norwegian bank–household relationship, we show that this cross-selling perspective amplifies monetary policy transmission: it reduces the pass-through of policy rate changes to deposit rates, so deposit and loan growth respond more strongly to monetary policy. Cross-selling thus extends bank market power to profits from multiple products and periods.

Keywords: monetary policy transmission, deposits channel of monetary policy, cross-selling, multi-product banking, bank franchise
JEL: D14, D43, E52, G21, G51

1. Introduction

The pass-through from monetary policy rates to deposit rates is typically incomplete: when policy rates rise, banks raise deposit rates less than one-for-one, widening the deposit spreads they earn. Because depositors can substitute into higher-pass-through alternatives—such as bonds or money market funds—this incomplete pass-through causes deposit volumes, and ultimately loan volumes, to respond to policy rate changes. The mechanism is central

*Philipp Schnabl was the editor for this article. This work should not be reported as representing the views of our former employers European Central Bank or Norges Bank. We are especially indebted to an anonymous referee and the editor, whose exceptionally careful readings and constructive comments across two rounds substantially improved the paper. For helpful comments or discussions we further thank, without implicating, participants of the 2024 Inaugural Conference of the Eurosystem Research Network "Challenges for Monetary Policy Transmission" (ChaMP), the 2023 Endless Summer Conference, the 2023 MoFiR conference, the ECB Research Brownbag, Federal Reserve Board, Yeshiva University, Bayes Business School, and the University of Zurich, as well as Juliane Begenau, Hans Degryse, Giovanni dell'Ariccia (discussant), Sebastian Doerr (discussant), Vasso Ioannidou, Luc Laeven, Camelia Minoiu (discussant), Federica Salvade (discussant), Alexi Savov, Glenn Schepens, Olivier Wang, and an anonymous Norges Bank referee. For assistance with additional analyses of euro area banks we thank Franziska Maruhn. Christoph Basten thanks the University of Zurich, the ECB's DG Monetary Policy and DG Research for opportunities to work on this project.

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to monetary policy transmission through the banking system. But what causes incomplete pass-through?

The leading explanation comes from the Deposits Channel framework (Drechsler et al., 2017), in which banks with deposit market power maximize the profits from their deposit relationships. When policy rates are higher, deposits become more attractive relative to zero-return cash, allowing banks to charge wider deposit spreads. The resulting incomplete pass-through generates higher deposit profits and reduces the sensitivity of deposit costs to policy rates, facilitating interest rate risk management (Drechsler et al., 2021).

Recent evidence, however, suggests that banks' deposit pricing may depend also on factors beyond the deposit product itself. Basten and Juelsrud (2026) show that banks may want to set deposit prices to maximize the net present value of each bank-client relationship, accounting also for future lending profits: because households tend to consolidate their banking relationships, a depositor today is a potential borrower tomorrow, and banks price deposits accordingly. That paper, however, concerns the cross-sectional *level* of deposit rates. What it leaves open—and what this paper is about—is how cross-selling affects the pass-through of policy rate *changes* to deposit rate changes: the pass-through at the heart of the deposits channel.

In this paper, we show that banks' expected ability to later cross-sell loans to current depositors, made possible by households' tendency to consolidate their banking relationships—*reduces* the pass-through of policy rate changes to deposit rates and thereby *magnifies* the deposits channel: deposit growth and loan growth respond more strongly to monetary policy where cross-selling potential is higher. Cross-selling thus reinforces the standard deposits channel in *changes* even though it works in the opposite direction of deposit market power in *levels*. We develop a model in which banks set deposit spreads to maximize the value of the entire customer relationship, linking the deposit and loan franchises that the standard framework treats as independent. We then test its predictions using administrative data covering the population of Norwegian bank-household relationships from 2004 to 2018. In our setting, cross-selling potential explains about as much of the heterogeneity in pass-through, deposit growth, and loan growth responses as traditional measures of deposit market concentration.

Our analysis proceeds in six steps. First, we develop a model that extends the Deposits Channel framework to incorporate cross-selling. When setting deposit spreads, banks trade off the marginal cost of compressing current deposit margins against the benefit of larger deposit volumes *and* the benefit of attracting depositors who may become profitable borrowers in the future; the optimal spread thus depends not only on deposit market power, as in the standard framework, but also on the net present value of expected cross-selling profits. Ex ante, the effect on pass-through is ambiguous. One intuition suggests that a bank competing for future borrowers would attract depositors with both higher deposit rate levels and higher pass-through of policy rate changes, which would dampen the deposits channel. Another suggests that clients who use more products are stickier, so cross-selling raises the bank's power over converted clients and lowers pass-through.

The model shows that these intuitions speak to different objects. The attraction motive governs the *level* of deposit rates: banks price deposits more attractively at every policy rate. Pass-through, by contrast, depends on how the *value* of that attractiveness varies with the policy rate. Because policy rate cuts raise the present value of future lending relationships—future profits are discounted less, and expected lending profits rise—banks offer the most

attractive deposit spreads precisely when policy rates are low. They consequently lower deposit rates by less when rates fall, and raise them by less when rates rise, than they would absent cross-selling: pass-through falls, and the responsiveness of both deposit growth and loan growth to monetary policy is amplified. Were the value of cross-selling invariant to the policy rate, it would shift deposit rate levels but leave pass-through unchanged. The lock-in intuition, where it operates, pushes in the same direction, though our empirical measure captures the prospective margin: the potential of current depositors to become future borrowers. Section 2 makes these points precise.

Second, we assemble administrative data covering the population of household depositors and borrowers in Norway from 2004 to 2018. These data include rich demographic, balance sheet, and income variables for every household—which is essential because cross-selling potential is significantly determined by demographics. Younger households, for instance, are substantially more likely to later take out mortgages from their deposit bank (Basten and Juelsrud, 2026). Using these observables, we estimate for each bank-household relationship the probability that the depositor will later borrow from the same bank (extensive margin), as well as the expected loan volume and spread (intensive margin). This allows us to examine how the same bank in the same year—and hence with identical refinancing needs—sets different pass-through rates for clients with different cross-selling potential.

Third, we document the effect of cross-selling on deposit pricing. Each standard deviation increase in cross-selling potential at the extensive margin is associated with up to 1 basis point lower pass-through; accounting also for expected loan volumes and spreads (the intensive margin), the effect roughly doubles. These magnitudes are comparable to the effect of a one-standard-deviation increase in deposit market concentration as measured by the HHI. Importantly, both survive jointly in the same specification, indicating that cross-selling and deposit market concentration capture complementary dimensions of market power.

A natural concern is that our measure of cross-selling potential, constructed from household demographics, could proxy for depositor sophistication or rate sensitivity rather than banks' forward-looking pricing incentives. We address this concern in two steps, developed in Section 4 and tested in Section 5.3. First, an Altonji–Oster comparison implies negative rather than positive selection: adding sophistication controls moves the coefficient away from zero, and a residualized measure of cross-selling potential purged of half of its variance still delivers a coefficient above the baseline. Second, a market-level decomposition establishes that the identifying variation is bank-municipality-specific rather than an attribute of the local area: other banks' cross-selling potential in the same municipality loses all explanatory power once market concentration is controlled for, while own-bank potential remains significant. Each step indicates that omitted variable bias, where present, attenuates rather than inflates the estimates.

Fourth, if cross-selling causally lowers pass-through, as our identification tests indicate, then banks with higher cross-selling potential offer relatively more attractive deposit rates when policy rates fall—and their depositors should respond accordingly. Consistent with this prediction, a one-standard-deviation increase in cross-selling potential amplifies the deposit growth response to policy rate changes by 0.7 to 1.4 percentage points in our preferred HHI-control specifications. As replacing deposits with alternative funding is typically costly, these deposit flow differences should also affect loan supply. Comparing corporate loan growth across banks with different cross-selling potential but within the same municipality-year

or sector-year—thereby holding loan demand fixed—we find effects of 0.6 to 1.9 percentage points per standard deviation, again comparable in magnitude to the effect of deposit market concentration.

Fifth, we decompose the cross-selling mechanism into two components. The more direct one operates through discounting: lower policy rates raise the present value of future lending profits, strengthening banks' incentive to attract depositors today. The second operates through flow profits: we find that policy rate cuts are also associated with higher expected loan volumes and wider loan spreads, reinforcing the first channel. While we can control for credit risk and rule out interest rate risk as drivers, we cannot fully isolate these flow-profit effects from demand-side changes, and so we view this decomposition as informative rather than definitive. Taken together, the evidence suggests that both components reinforce each other, amplifying the overall effect of cross-selling on pass-through.

Finally, since the cross-selling mechanism—like the deposits channel itself—incentivizes lower deposit spreads at lower policy rates and may therefore at some point make the deposit business on its own loss-making, we ask in Section 5.7 whether it has an effective lower bound. The average deposit spread falls to zero once the policy rate reaches about 1.5 percentage points and turns negative below it (Figure 1); in the years in which policy rates fell below 1.5 percentage points, the link between cross-selling potential and pass-through weakens sharply, and increasingly so for banks with fewer financial resources. Cross-selling incentives thus do not vanish at low policy rates, but banks' capacity to act on them becomes increasingly dependent on their financial strength.

Our primary contribution is to the literature on the Deposits Channel of monetary policy (Drechsler et al., 2017, 2021, 2023; Granja and Paixão, 2026). We do not view cross-selling as a separate transmission mechanism but rather as a previously underexplored dimension of bank market power. Where the existing literature measures deposit market power through local market concentration, we show that banks' ability to convert depositors into future borrowers constitutes a dynamic, relationship-level form of market power that independently *reduces* the pass-through of policy rate changes to deposit rates and *amplifies* the deposit growth and loan growth responses to monetary policy, thereby magnifying the standard deposits channel. This broadens the concept of deposit market power central to the Deposits Channel—beyond static concentration to also encompass client stickiness across products and time—and generates heterogeneity in deposit betas that concentration alone does not capture. In this sense, we expand the notion of the deposit relationship in Drechsler et al. (2017)—in which the bank maximizes the income from that relationship—to encompass the profits from the broader customer relationship, including future lending.

The paper closest to ours is Basten and Juelsrud (2026), which established that cross-selling potential affects deposit pricing in the cross-section. We contribute relative to that paper by asking a different question: not whether cross-selling affects the *level* of deposit pricing, but whether it affects the *pass-through* of monetary policy changes—and trace the implications for deposit and loan growth dynamics.

Second, we contribute to the industrial organization literature on loss-leader pricing and customer acquisition (Klemperer, 1995; Carbo-Valverde et al., 2011). While the theoretical logic of acquiring customers at below-cost prices to extract follow-on profits is well established, we show that the strength of incentives for loss-leader pricing may vary with macroeconomic conditions.

Third, we contribute to the literature strands on bank franchise value (Drechsler et al., 2021; Bolton et al., 2025; DeMarzo et al., 2024; Narayanan et al., 2025) and multi-product banking or cross-selling in banking (Basten and Juelsrud, 2026; Qi, 2024). A common feature of existing models is that deposits matter for loans only through funding: banks raise deposits to finance lending. We show that the link is richer—the deposit franchise also serves as a pipeline for future lending relationships. In the Monti-Klein framework (Klein, 1971; Monti, 1972), optimal deposit and loan quantities are determined independently; in our setting, they are jointly optimized through the customer relationship, creating an additional channel through which deposit market conditions influence loan supply.

The paper proceeds as follows. Section 2 develops the conceptual framework. Section 3 describes the data. Section 4 outlines the empirical strategy. Section 5 presents results: first on deposit pricing and its causal identification, then on deposit and loan growth, and finally on heterogeneity across bank types and policy rate environments. Section 6 concludes.

2. Conceptual Framework

In this section, we develop a simple model showing how cross-selling potential affects deposit pricing and derive the testable predictions we take to the data.

Consider a two-period model with periods "today" and "tomorrow." A monopolistically competitive bank sets a deposit rate r_d, which determines the deposit spread $s_d = r - r_d$ relative to the exogenous policy rate r. The bank faces a strictly downward-sloping deposit demand curve $D(s_d)$ today.

At the same time, greater deposit volumes today are associated with greater loan volumes tomorrow, motivated by the finding in Basten and Juelsrud (2026) whereby an existing deposit relationship makes a client 20 percentage points more likely to also borrow from the same bank later compared to an otherwise identical client and an otherwise identical bank but with no existing deposit relationship. Additional regressions confirm that this positive extensive margin effect from deposit to loan relationships translates also into a positive (unconditional) intensive margin effect whereby higher deposit volumes are also associated with in expectation higher future loan volumes and so we assume in our simple conceptual framework that L is an increasing function of deposit volume D: a lower deposit spread today then attracts not only more deposits today but also more loans tomorrow.[1] Denoting loan volume as L and the loan spread (loan rate minus policy rate) as l, future cross-selling profits are then $L \times l$.

With cross-selling, deposit spreads set today influence both current deposit profits and the flow profits from cross-selling tomorrow. The latter are discounted at the rate $\frac{1}{1+r}$ to obtain their net present value. At baseline, we assume that the bank internalizes tomorrow's cross-selling profits when setting today's deposit rate, discounting them at the policy rate r. We discuss below how the predictions change when decision-makers are more myopic—for instance, due to short-term compensation incentives. We assume risk neutrality for

[1]Basten and Juelsrud (2026) show also cross-selling of future deposits to current depositors, or depositor stickiness, but at least in our empirical setup cross-selling of loans yields on average both greater volumes and higher spreads. Therefore, we omit here the term for profits from future deposits but note that it would not change our key analyses.

tractability. Finally, we allow both the loan spread l and loan volume L to vary with the current policy rate r, and investigate empirically whether they do in Section 5.

The bank's optimization problem is:

$$\max_{r_d} (r - r_d) \times D(r - r_d) + \frac{1}{1+r} l(r) \cdot L(D(r - r_d)) \tag{1}$$

The first term is the standard deposit profit function under monopolistic competition (see, e.g., Freixas and Rochet, 2008); the second is the discounted value of expected cross-selling profits. For brevity, we suppress the expectations operator throughout, but all cross-selling terms should be understood as expected values.[2]

The net present value of cross-selling depends on the policy rate through two channels. The first is discounting: since the bank is risk-neutral, future cross-selling profits are discounted at the policy rate r. In practice, cross-selling profits in our setup materialize on average five years after the deposit relationship begins (Basten and Juelsrud, 2026), so the relevant discount rate may correspond to a longer maturity. However, overnight and longer-term rates are positively correlated, and Figure 2 confirms that incomplete pass-through obtains also when we benchmark against the 3-month money-market rate or the 10-year government bond yield. The second channel operates through flow profits: if the loan spread $l(r)$ or loan volume L also varies with r, then policy rate changes affect not only the discounting of future profits but their magnitude as well. We investigate this possibility empirically in Section 5.

The bank trades off the marginal cost of a lower deposit spread against two marginal benefits: greater deposit volume today, and greater expected cross-selling profits tomorrow. Assuming constant elasticity of deposit demand, $\epsilon_D \equiv -D'(r - r_d) \times \frac{r_d}{D(r-r_d)} > 0$, the optimal deposit rate is:

$$r_d = \frac{1}{\mu} \left(r + \frac{1}{(1+r)} l(r) L'(D(r - r_d)) \right) \tag{2}$$

Here $\frac{1}{\mu} \equiv \frac{\epsilon_d}{\epsilon_d + 1}$ is the deposit markup, capturing market power in the deposit market alone, while $L'(D)$ captures cross-selling potential—the sensitivity of future loan volume to current deposit volume. From this expression, we derive the deposit spread beta, defined as the marginal response of the deposit spread to a change in the policy rate.

Proposition 1. *(Deposit market power and cross-selling potential both dampen pass-through.) The deposit spread beta is given by*

$$DSB \equiv \frac{\partial s}{\partial r} = 1 - \frac{\partial r_d}{\partial r} = 1 - \frac{1}{\mu} \left(1 - \frac{1}{1+r} L'(D(r - r_d)) \left(\frac{1}{1+r} l(r) - l'(r) \right) \right) \tag{3}$$

and is

[2] An alternative way to formulate this would be to express the customer base, part of a firm's intangible capital, as a state variable in a setup where clients have search costs so that the bank, or any firm, will invest into client acquisition to the extent to which the expected net present value of future profits warrants it, as in Gourio and Rudanko (2014).

a. *increasing in deposit market power μ (equivalently, decreasing in the elasticity of deposit demand ϵ_D);*
b. *increasing in cross-selling potential $L'(D)$, provided the discounting effect dominates the flow-profit effect, i.e., $\frac{1}{1+r}l(r) > l'(r)$.*

PROOF. Applying the Implicit Function Theorem to Equation (2).

The condition in part (b) requires that the increase in the present value of cross-selling from a lower policy rate is not fully offset by a decline in loan spreads. Empirically, we find that loan spreads decrease when policy rates rise ($l'(r) < 0$), so both effects reinforce each other and the condition holds comfortably.[3]

The model separates the effect of cross-selling on the *level* of deposit rates from its effect on *pass-through*—the two go in opposite directions relative to deposit market power. In Equation (2), the cross-selling term $\frac{1}{1+r}l(r)L'(D)$ raises the optimal deposit rate at every policy rate: the attraction and retention motive makes the bank price deposits more attractively in levels, working against deposit market power, as documented in Basten and Juelsrud (2026). Pass-through, however, is governed not by the size of this term but by its *derivative* with respect to the policy rate. Were the net present value of cross-selling invariant to r, the term would shift deposit rate levels while leaving $\partial r_d/\partial r = 1/\mu$—and hence pass-through—unchanged. Because that value instead falls as the policy rate rises, through discounting and, given $l'(r) < 0$, through flow profits, the bank's deposit attractiveness is countercyclical in the policy rate: it lowers deposit rates by less when rates fall, and raises them by less when rates rise, than it would absent cross-selling. This is why cross-selling reduces pass-through and reinforces the deposits channel in changes even as it opposes it in levels.

A complementary force, not modeled here, points the same way: a deposit account tied to a mortgage is costly to switch, so cross-selling also locks converted clients in, lowering their deposit demand elasticity ϵ_D and raising μ. We hold ϵ_D fixed across clients to isolate the cross-selling-value channel, and note that our empirical measure captures the prospective margin—the potential of current depositors to become borrowers—while lock-in of already-converted clients would further reduce pass-through in the same direction.

The loan spread $l(r)$ plays an important role in the model, as it determines how profitable cross-selling of loans is. In the setup we study, loan spreads are, on average through the cycle, empirically larger than deposit spreads, so that the primary cross-selling objective of most client onboarding is arguably the cross-selling of future loans. In other settings, the attraction and retention of current deposit clients could equally be motivated by valuable future deposit profits expected for when policy rates are higher again, or by future profits from cross-selling other banking products such as wealth management services, payment services, or insurance.

[3] A crucial assumption is that the decision-makers who set deposit prices are sufficiently forward-looking to internalize future cross-selling profits. This could fail if, for instance, short-term compensation structures or career concerns lead managers to overweight current deposit profits relative to future lending relationships. Our empirical finding that cross-selling potential significantly predicts deposit pricing is consistent with at least the average decision-maker being forward-looking, though we cannot rule out all alternative interpretations—a point we address through our identification strategy in Section 5.

In that sense expected future margins on other products determine which products a bank may aim to cross-sell and thereby which type of client it wants to attract now.

We test the relevance of loan market power empirically using the loan-market municipal Herfindahl-Hirschmann Index. We find that loan HHI to be highly correlated with deposit-market HHI (approximately 0.90), as highlighted by Figure A.1, confirming that banks that dominate local deposit markets also dominate local loan markets. And when we add loan-market HHI interacted with the policy rate change to our baseline regression, it significantly dampens pass-through, consistent with the idea that a more profitable loan business makes the cross-selling of loans all the more attractive (Table A.3). Given the 0.90 correlation between the two HHIs, the separate coefficients on loan- and deposit-market concentration in that table are difficult to interpret. The more informative fact is that the coefficient on cross-selling potential is essentially unchanged when loan-market HHI is added: our measure picks up a dimension of loan market power distinct from local concentration, namely the relationship-level potential to convert depositors into borrowers.

Finally, because deposit demand $D(s_d)$ is downward-sloping, the effect of cross-selling on deposit spreads carries over to deposit quantities: banks with higher cross-selling potential exhibit not only pass-through but also greater sensitivity of deposit growth to policy rate changes. And because deposits are banks' primary source of loan funding, more responsive deposit growth should translate into more responsive loan growth. We test these predictions—on deposit pricing, deposit growth, and loan growth—in the remainder of the paper.[4]

3. Data and Summary Statistics

This section describes the data sources, defines key variables, and presents summary statistics.

3.1. Data Sources and Preparation

Our dataset combines three sources; Table 1 reports summary statistics for the merged sample.

Deposit and loan accounts. The first source is administrative data on every bank-household deposit and loan relationship in Norway for 2004–2018. The data are reported annually for tax purposes by both households and banks, and cover outstanding balances and *effective* interest rates at the relationship level, not just posted rates. This granularity is central to our identification strategy: it allows us to compare how the same bank in the same year prices deposits differently across clients with different cross-selling potential. Roughly 85% of household loans are mortgages, and over 99% of deposits have a contractual maturity of less than one year.

[4]Greenwald et al. (2023) recently showed that in practice deposit betas may in turn be increasing in market rates. This could more strongly reduce banks' incentives to lend or weaken the increase in deposit spread betas and hence the reduction in deposit and resulting loan volumes in the Drechsler et al. (2017) lens, thus either strengthening or weakening monetary policy transmission. Given our focus on cross-selling incentives rather than deposit profits alone, we ignore this here although different estimates for different policy rate levels between 50bp and 500bp would be a possible extension of our conceptual framework.

Other household information. The second source is individual-level tax records covering all Norwegian taxpayers, observed annually. These include balance sheets, income statements, and demographic variables: age, postcode, education, and child benefit receipt. These demographics serve dual roles: they are inputs to our cross-selling potential estimates and, in robustness tests, proxies for depositor sophistication (Section 5).

Bank-level information. Third, we add bank-level balance sheet and income statement data from the ORBOF supervisory database, which covers all banks operating in Norway.

Combining the data. All three sources share individual and bank identifiers, allowing exact matching. We aggregate individual-level data to the household level, then to bank-municipality-year cells, the unit of analysis for our baseline regressions. Section 3.2 defines the key variables; Section 4 explains why this level of aggregation is appropriate.

3.2. Variable Construction

Cross-selling indicators. Our baseline measure of cross-selling potential is the estimated probability that an existing depositor later borrows from the same bank. We estimate this using a linear probability model on the subsample of bank-household-year observations with positive deposits, where the dependent variable is an indicator equal to one if the household subsequently takes a loan from the same bank during our sample period. Restricting to existing depositors aligns the estimation sample with the object the cross-selling channel concerns: the value of converting a current deposit relationship into a future lending relationship. The predictors are household demographics and relationship characteristics that the bank can also observe: age, parenthood, retirement status, income, wealth, deposit balances, number of existing bank relationships, and local market structure. Table 2 reports the full results.

Several predictors stand out. Households headed by someone under 30 are 14.5 percentage points more likely to convert from depositor to borrower and are associated with substantially higher expected loan volumes. Parenthood reduces conversion probability but, because of larger loan volumes conditional on borrowing, is associated with higher expected loan profits. Both youth and parenthood are associated with *lower* loan spreads, which could moderate a client's attractiveness to the bank. Each additional banking relationship a household maintains is associated with lower conversion probability, while more concentrated local markets are associated with higher propensity.

Because the net present value of cross-selling depends not only on whether a depositor borrows but also on how much and at what spread, we also estimate expected loan volume and expected loan spread using the same observables. Borrowing propensity is our baseline measure due to its transparent interpretation; the volume and spread measures serve as alternatives that capture the intensive margin. We prefer the estimated propensity over the raw fraction of depositors who borrow within our sample, since the estimated measure also captures borrowing that demographics predict but that occurs after the sample period ends.

Measures of deposit market concentration. Given the central role of deposit market concentration in the existing literature, we also construct standard measures for comparison. Our primary concentration measure is the Herfindahl-Hirschman Index (HHI), computed as the sum of squared deposit market shares within each municipality. As a more granular alternative, we compute each bank's own deposit market share in each municipality, which, unlike HHI, varies across banks within the same municipality.

Deposit spreads and aggregation. We define the deposit spread as the policy rate minus the effective deposit rate, where the latter is computed as deposit interest earned divided by the average of current- and prior-year deposit volumes. Averaging over two years mitigates downward bias when deposits are added late in the year.

We aggregate from individual bank-household relationships to bank-municipality-year cells, computing mean deposit spreads and summing deposit volumes before calculating growth rates. This aggregation serves two purposes. First, it smooths the extreme outliers in household-level deposit growth that arise from near-zero initial balances. Second, it aligns with our treatment of municipalities as the relevant geographic market: we consider a household able to bank with any institution that already has at least one client in that municipality. In most cases this coincides with branch presence, though some relationships are maintained through online banking after a household relocates.

3.3. Summary Statistics

Table 1 reports summary statistics. Panel A describes the roughly 45 million bank-household-year observations; Panel B aggregates to our baseline bank-municipality-year level. Panels C1–C3 cover the loan growth samples at the bank-municipality, bank-NACE3, and bank-NACE5 levels, and Panel D presents bank-level aggregates used in the descriptive analysis of Section 5.1.

Over our sample period, the Norwegian policy rate ranged from 0.5% to 5.3%, with year-on-year changes of between -3.6 and $+1.6$ percentage points. Average deposit spreads are approximately 0.5 percentage points at both the relationship and bank-municipality levels (Panels A and B); Figure 1 plots the volume-weighted average spread over time, which compresses as policy rates decline and is lowest in the post-2015 low-rate years that we examine separately in Section 5.7. Mean deposit volumes are NOK 135,000 (approximately USD 13,500) per relationship and NOK 22 million per bank-municipality cell.

Turning to the key explanatory variables, the average estimated probability that an existing depositor later borrows from the same bank (PrB) is 17.8% at the bank-municipality level, meaning that roughly one in six deposit relationships eventually generates a loan from the same bank. HHI averages 0.29 (relationship level) to 0.38 (bank-municipality level), indicating moderately concentrated local deposit markets. Bank-level market shares average 21% at the relationship level and 1.6% at the bank-municipality level, reflecting the large number of municipalities over which most banks operate.

In the loan growth samples (Panels C1–C3), average corporate loan growth ranges from -2.6% to -6.5% depending on the level of aggregation and computation method. HHI averages approximately 0.3 and borrowing propensity approximately 0.2 in these samples.

4. Empirical Strategy

This section describes our empirical approach. We first present the baseline specification for deposit pricing, which exploits within-bank-year variation across municipalities, and discuss how we address potential confounding between cross-selling potential and depositor sophistication. We then extend the framework to deposit growth and loan growth.

4.1. Analyzing Deposit Pricing

Our initial goal is to estimate how the deposit spread beta—the effect of policy rate changes on deposit spread changes—varies with cross-selling potential. Two identification concerns arise. The first is that policy rate changes affect loan demand, which in turn affects banks' refinancing needs and deposit pricing. Because these demand shifts may vary across banks, they could confound the relationship between cross-selling potential and pass-through. We address this through bank-year fixed effects, which absorb all bank-level time-varying factors, including refinancing needs, and identify the cross-selling effect from variation across municipalities within the same bank-year.

The second concern is that our measure of cross-selling potential, constructed from household demographics, could proxy for observable or unobservable determinants of interest rate pass-through other than cross-selling potential, such as depositor sophistication or rate sensitivity. We address this in two steps. First, an Altonji–Oster analysis and a residualization exercise indicate that selection, whether on observables or unobservables, if anything biases the estimates down- rather than upward. And second, we confirm that our baseline estimates rely on variation in cross-selling potential unique to each bank-municipality combination rather than variation common to all cells in the same municipality, which could also have been driven by other municipality-level factors. The remainder of this section details each step; Section 5.3 reports the results.

Consider the first concern. Different policy rate levels may be associated with different loan demand, either because central banks raise rates in response to credit booms or because rate increases directly reduce borrowing. These demand shifts affect banks' refinancing needs, and hence their deposit pricing. The standard solution, introduced by Drechsler et al. (2017), is to compare deposit pricing across locations within the same bank and year. Because deposits raised locally are pooled at the bank level to fund lending, bank-year fixed effects absorb refinancing needs and isolate local variation in deposit pricing.

However, the within-bank-year approach has limitations in the US context. Granja and Paixão (2026) document that many US banks set deposit rates uniformly across their branch networks, which may limit the strategy's external validity to the subset of banks that do price locally. Narayanan et al. (2025) take a different approach, predicting branch-level deposit franchise values from local demographics and examining how branch openings and closures vary with these predicted values within each bank-year. A common insight across these papers is that *effective* deposit pricing varies with client demographics even when posted rates do not, a feature our data allow us to exploit directly.

Our data and institutional setting offer three advantages over the US branch-level approach. First, we observe deposit pricing at the bank-household level, finer than the branch level, while controlling for refinancing needs through bank-year fixed effects. Second, we observe *effective* deposit rates for each relationship, not just posted rates. In practice, Norwegian banks do not set household-specific deposit rates, but they do offer multiple deposit products, some with demographic eligibility criteria, and pass-through varies across products. Because households sort into different products, effective rates and pass-through differ across households within the same bank-year (Basten and Juelsrud, 2026). This variation captures differences in the bank's effective market power over each client, regardless of whether it originates from product design or household selection. To quantify this variation, we compute the standard deviation of effective deposit rates across households within each bank-year. The

mean within-bank-year standard deviation is 1.03 percentage points (median: 1.00), with even the 25th percentile at 0.82 percentage points, and this dispersion is stable across bank-years (its cross-bank-year standard deviation is only 0.25 percentage points). For reference, Basten and Juelsrud (2026, Table 1) report an overall standard deviation of effective deposit rates of 1.18 percentage points across bank-household-year observations. The within-bank-year component therefore accounts for the large majority of total rate dispersion, confirming that effective deposit rates vary substantially across households served by the same bank in the same year, providing ample scope for the cross-selling mechanism to operate through differential pricing. Third, bank-level pooling of deposit funding is plausible in our setting: there are no regulatory restrictions on banks' internal capital markets, and Norwegian banks are geographically diversified, with the median bank operating across 340 municipalities and even the 5th percentile at 42.

As discussed in Section 3, we aggregate from bank-household relationships to bank-municipality-year cells to avoid the noise in household-level deposit growth rates. The baseline specification is:

$$Y_{b,m,y} = \alpha + \beta_1 \Delta PolRate_y \times PrB_{b,m,y} + \beta_2 \Delta PolRate_y \times HHI_m + \ldots + \delta_{b,y} + \epsilon_{b,m,y} \quad (4)$$

where b indexes the bank, m the municipality, and y the year. The dependent variable Y is the change in deposit spreads or deposit growth. Policy rate changes $\Delta PolRate_y$ are computed year-on-year to match our annual data frequency. PrB is our baseline measure of cross-selling potential (borrowing propensity, averaged across households in each bank-municipality cell), and HHI measures deposit market concentration; some specifications replace HHI with the bank's own market share MS. The omission dots in the equation capture the non-interacted levels of these variables. The coefficient of interest is β_1, which captures how much more deposit spreads respond to policy rate changes in bank-municipality cells with higher cross-selling potential. The bank-year fixed effects $\delta_{b,y}$ absorb all bank-level time-varying factors, including refinancing needs, so identification comes from comparing municipalities with different cross-selling potential within the same bank-year. This within-bank comparison also accounts for bank sorting across municipality types (d'Avernas et al., 2023).

We now turn to the second concern. We estimate our PrB measure of cross-selling potential from household demographics, including age, income, and wealth. These same characteristics plausibly correlate with drivers of rate level or pass-through sensitivity other than cross-selling potential, in which case our coefficients on PrB might partially pick up the effect of those drivers rather than those of cross-selling potential itself. A related question is which aspect of loan market power PrB captures. It is not local loan market concentration: adding loan-market HHI, interacted with the policy rate change, to the baseline leaves the PrB coefficient essentially unchanged (Table A.3), consistent with PrB picking up relationship-level conversion potential rather than market structure, with which deposit-market HHI is in any case nearly collinear (correlation 0.90). We address the composition concerns in two steps.

First, we formalize robustness to observables and unobservables. Following Altonji et al. (2005) and Oster (2019), we compare the coefficient in a short specification with bank-year fixed effects only against a long specification adding interactions of the policy rate

change with four proxies for depositor sophistication—education, residential mobility, stock-market participation, and the share of wealth held in stocks and bonds—which are not inputs to PrB and so isolate the independent variation the concern is about; movement of the coefficient toward zero would indicate positive selection on observables, movement away from zero negative selection. As the most flexible variant of the same logic, we residualize PrB within bank-year against the full control set—depositor demographics (age shares for five brackets), financial characteristics (average log income and average log wealth), and the four sophistication proxies—and use the residual, purged by construction of all linear composition effects including the demographic ingredients of PrB itself, as the regressor.

Separately, we also examine the timing of the pass-through response by adding leads and lags of the policy-rate change to the baseline specification. Because depositor demand and bank incentives respond to the same contemporaneous rate change, this exercise speaks to dynamics and anticipation rather than to the sophistication concern, and we report it with the robustness checks.

Second, we note that cross-selling potential in a cell can be decomposed into a municipality(-year) component common to all banks in the market and be specific to certain banks in that municipality(-year). If the coefficient on cross-selling potential reflected an attribute of the area rather than of the bank—local demographics, local rate sensitivity, or any other condition every bank in the market faces—the leave-one-out average of other banks' PrB in the same municipality-year should predict a bank's pass-through in that municipality as well as its own PrB there does. We estimate joint specifications with both components, with and without deposit market concentration.

4.2. Analyzing Deposit Growth

We also estimate Equation (4) with deposit growth as the dependent variable. The logic is straightforward: if cross-selling causally lowers pass-through, as our identification tests for deposit pricing suggest, then banks with higher cross-selling potential offer relatively more attractive deposit rates when policy rates fall, and their depositors should respond with larger deposit inflows. The deposit growth regressions test this implication rather than independently establishing a causal effect of cross-selling on deposit quantities. The identifying assumption parallels the pricing specification: conditional on bank-year fixed effects, differential deposit growth across municipalities within the same bank reflects depositors' responses to differential pricing rather than local economic shocks. As a robustness check, we replace bank-year with municipality-year fixed effects to absorb time-varying local factors, and also control directly for local unemployment and house price growth.

4.3. Analyzing Loan Supply

The same logic extends to loan growth: if cross-selling affects deposit pricing and deposit growth, it should ultimately affect banks' lending capacity. However, the identification challenge is harder. Bank-year fixed effects, which absorb refinancing needs in the deposit pricing regressions, no longer suffice because the variation of interest, loan growth, now occurs at the bank level. Policy rate changes may independently shift loan demand, and if loan demand correlates with cross-selling potential, the estimates would be confounded.

Following Drechsler et al. (2017), we address this by comparing loan growth across banks within the same municipality-year, so that all banks serving a given local market face the

same loan demand conditions. Cross-selling potential is now measured at the bank level (averaged across all of a bank's municipality cells), and municipality-year fixed effects replace bank-year fixed effects.

We focus on loans to non-financial corporations (NFCs) rather than household loans. This is important because our cross-selling measure is based on household mortgage potential. By examining corporate lending, we ensure that the loan growth results reflect the indirect effect of differential deposit funding on banks' overall lending capacity, not a mechanical link between depositors becoming borrowers.

The loan growth sample comprises 178 banks offering both household deposits and corporate loans, active across up to 491 municipalities, with an average of 37 banks per municipality.

$$Y_{b,m,y} = \alpha + \beta_1 \Delta PolRate_y \times PrB_{b,y} + \beta_2 \Delta PolRate_y \times HHI_m + \ldots + \delta_{m,y} + \epsilon_{b,m,y} \quad (5)$$

where Y is loan growth (log-difference or symmetric growth) or the log volume of new loans by bank b in municipality m in year y. The key differences from Equation (4) are that PrB is measured at the bank level and municipality-year fixed effects $\delta_{m,y}$ replace bank-year fixed effects, as described above.

As a robustness check, we also estimate loan growth at the bank-sector level using NACE 3-digit industry codes, with sector-year fixed effects replacing municipality-year fixed effects. This controls for loan demand at the industry level rather than the geographic level. Consistent results across both approaches would reinforce the interpretation that cross-selling potential affects loan supply through the deposit funding channel established in the pricing and deposit growth analyses.

5. Results

We begin with descriptive evidence linking bank-level pass-through to cross-selling potential (Section 5.1), then present the core pricing results (Section 5.2) and address potential confounds (Section 5.3). Sections 5.4 and 5.5 trace the implications for deposit growth and loan growth. Section 5.6 decomposes the mechanism into discounting and flow-profit subchannels, Section 5.7 examines heterogeneity across bank types and policy rate environments, and Sections 5.8 and 5.9 present additional robustness checks and external validity evidence.

5.1. Descriptive Evidence

We begin by estimating pass-through separately for each bank. Figure 3 shows that median annual pass-through is about 14% and mean pass-through about 15%, with substantial heterogeneity across banks even after winsorizing the top and bottom 5% of the distribution.

Figure 4 relates this heterogeneity to cross-selling potential. The bin scatter plots bank-level deposit spread betas against average borrowing propensity. Across banks, regressing the beta on the bank's average PrB yields a slope of -0.95 (SE $= 0.41$): a bank whose depositors are 10 percentage points more likely to later borrow exhibits roughly 9.5 percentage points lower pass-through. The formal within-bank evidence, including how the relationship varies with the rate environment, follows in Sections 5.2–5.7.

5.2. Deposit Pricing

Our baseline sample contains approximately 71,000 bank-municipality pairs observed over up to 15 years, yielding roughly 360,000 observations. Table 3 reports the core pricing results. Higher cross-selling potential is associated with significantly higher deposit spread betas: the coefficient on $PrB \times \Delta PolRate_y$ ranges from 0.10 to 0.18 across specifications (columns 2–4), meaning that a bank-municipality cell where all depositors are certain to convert from depositor to borrower exhibits a 10 to 18 percentage points higher deposit spread beta—that is, 10 to 18 percentage points lower pass-through—than one with no cross-selling prospect. Controlling for deposit market concentration (HHI) reduces the cross-selling coefficient by about 40%, but both variables remain economically and statistically significant, confirming that cross-selling and deposit market concentration both contribute independently to explaining heterogeneity in pass-through.

To compare magnitudes, Table 3 also reports effects per standard deviation. A one-SD increase in cross-selling potential raises the deposit spread beta by 0.5 to 0.9 basis points, comparable to the 0.7 basis point effect of a one-SD increase in HHI. Bank-level market share has no significant effect. The cross-selling magnitude is similar to the HHI effect reported by Drechsler et al. (2017) for US banks.

The un-interacted PrB term is negative and significant. This implies that the general inter-temporal compression of deposit spreads in the sample period, which was largely characterized by a general consolidation of the banking sector in addition to the low-for-long policy rates, was within the average bank more pronounced in municipalities with greater than in municipalities with smaller cross-selling potential, even when averaging across episodes with different policy rate changes. This is consistent with the idea that the average bank in our sample managed to strengthen the cross-sectional loss-leader pricing observed in Basten and Juelsrud (2026) over time, as it accumulated better data and experience on profitable cross-selling clients. The interaction $PrB \times \Delta PolRate_y$ then captures how much more this incentive strengthens when policy rates fall.

Table 4 replaces borrowing propensity with the intensive-margin measures defined in Section 3. Using expected loan volume, the effect per standard deviation ranges from 0.5 to 0.8 basis points, similar to the baseline. Using expected loan income, which accounts for both volume and spread, the effect rises to 1.4 to 1.6 basis points, roughly twice the volume-based effect. This suggests that banks price in not just the probability of cross-selling but the expected profitability of the future lending relationship. Expected loan income is arguably the object a profit-maximizing bank ultimately cares about; we retain borrowing propensity as the baseline for its transparent interpretation, and read the income-based estimates as the upper end of the economically relevant range.

5.3. Addressing Potential Confounds

Before tracing the implications of these pricing results for deposit and loan growth, we examine whether the estimated cross-selling effect could be driven by confounding from depositor sophistication, proceeding in two steps as discussed in Section 4.

Robustness to observables and unobservables: Altonji–Oster and residualization. Table A.1 compares a short specification (bank-year fixed effects only) with a long specification adding the four sophistication interactions. The coefficient *increases* from 0.166 (SE = 0.032)

to 0.215 (SE = 0.036) while the R-squared is nearly unchanged (0.2648 to 0.2664). Movement away from zero implies negative selection: selection on unobservables proportional to selection on observables would amplify, not eliminate, the effect. We deliberately do not report a numeric Oster bound; with the near-zero change in R-squared it is numerically unstable, and the informative statistic is the coefficient's stability. Columns 3–4 take the same logic further by orthogonalizing PrB within bank-year against age, income, and wealth (column 3) and additionally against the sophistication proxies (column 4), using the residual as the regressor. Bank-year fixed effects and the controls jointly absorb 51% of the variance of PrB—17 percentage points from the fixed effects and 34 from the controls, of which the demographic block accounts for nearly all (a within-bank-year partial R-squared of 0.40, against 0.016 for the sophistication proxies)—yet the residual still carries a coefficient of 0.217 (SE = 0.042), above the baseline coefficient of 0.166.

Bank-specific potential versus area conditions. PrB in a cell can be decomposed into a municipality-year component common to all banks in the market and a bank-specific deviation therein. If our estimates reflected an area attribute—local demographics, local rate sensitivity, or any other condition every bank in the market faces—the market component should predict pass-through as well as the bank's own PrB does. Table 5 tests this with the leave-one-out average PrB of all other banks in the same municipality-year. Once deposit market concentration (HHI $\times\Delta$PR)—the control we carry throughout the paper—is added, the other-banks term is zero (0.005, SE = 0.118) while own-bank PrB remains significant (0.102, SE = 0.037). The market-level component is absorbed by market structure; what identifies the coefficient is bank-municipality-specific cross-selling potential, rejecting even further the concern that it might simply reflect other drivers, such as average age or education, varying at the municipality level.

Taken together, both parts point the same way: omitted variable bias, where present, attenuates rather than inflates our estimates.

5.4. Deposit Growth

Given the pricing effects established above, banks with higher cross-selling potential should offer relatively more attractive deposit rates when policy rates fall, generating larger deposit inflows. Table 6 tests this prediction using deposit growth (log-difference) as the dependent variable. We focus on the specification that includes deposit market concentration (HHI $\times\Delta$PR), consistent with the joint specification used throughout the pricing analysis. In this specification a one-SD increase in cross-selling potential amplifies the deposit growth response by roughly 0.7 to 1.4 percentage points, and the interaction is statistically significant only once HHI is included (columns 3 and 7 of Table 6); in the specifications without HHI the effect is negative in five of six cases but not significant at conventional levels. Deposit growth is well known to be noisy at the bank-municipality level, which is reflected in the precision of these estimates. We report the result transparently: the deposit-growth channel points in the direction the model predicts and is statistically significant in the HHI-control specification that is the natural complement to the pricing regressions in Table 3, but it is less robustly estimated than the pricing result itself.

5.5. Loan Growth

Because deposits cannot easily be replaced with alternative funding, differential deposit growth should translate into differential loan supply. Table 7 tests this using NFC loan growth at the bank-municipality level, with municipality-year fixed effects to absorb loan demand (as described in Section 4). A one-SD increase in cross-selling potential amplifies the loan growth response by approximately 1.6 percentage points per 100 basis point rate change (1.9 in the joint specification with HHI), comparable to the effect of a one-SD increase in HHI (1.1 percentage points). When both enter jointly, the cross-selling effect strengthens while HHI becomes insignificant. Results are qualitatively similar using symmetric growth rates and when measuring the volume of new lending.

Table 8 confirms these patterns at the bank-sector level using NACE 3-digit industry codes, with sector-year fixed effects replacing municipality-year fixed effects. The stand-alone magnitude is smaller (0.6 percentage points per SD), while the joint specification with HHI yields 1.8 percentage points, close to the bank-municipality estimate. Cross-selling potential remains significant alongside HHI in the joint specifications, although the HHI interaction there takes the opposite sign to the bank-municipality level. Table A.4 reports similar results at the NACE 5-digit level.

The loan growth effects (0.6–1.9 percentage points per SD) are of comparable magnitude to the deposit growth effects (0.7–1.4 percentage points per SD in the preferred HHI-control specifications), consistent with deposit funding shocks transmitting through to lending capacity.

5.6. Discounting and Flow-Profit Sub-Channels

The model identifies two sub-channels through which policy rates affect the value of cross-selling. Lower rates raise the present value of future profits (discounting) and may also raise the profits themselves if loan volumes or spreads increase (flow profits). We examine the second channel using bank-household-level data that allow us to control for individual loan credit risk.

Table 9 reports the results, restricting the sample to relationships that ever held deposits to align with the cross-selling channel of interest. A 100 basis point policy rate increase is associated with economically significant reductions, across the specifications of both panels, in the probability that a depositor later borrows (2.9 to 5.5 percentage points), in expected loan volume (5.9 to 15.5 percentage points), and in loan spreads (22 to 28 basis points); Figure 5 shows the loan-spread relationship visually. The conversion probability of existing depositors responds sharply, consistent with the cross-selling mechanism operating through the conversion of existing depositors into borrowers.

A natural concern is that lower loan spreads at lower policy rates could reflect lower credit risk rather than higher bank profitability. We construct proxies for non-performing loans and find that they carry a significant risk premium (22 to 48 basis points; Table A.5). Controlling for later NPL status leaves the loan-spread response essentially unchanged, while the responses of the conversion probability and of loan volume shrink in the unweighted panel (columns 7–9 of Table 9); the spread-based component of the flow-profit evidence is therefore the most robust. Since nearly all loans in our sample carry adjustable rates (Basten and Juelsrud, 2026), interest rate risk is also unlikely to drive the spread variation. We note, however, that we cannot fully separate supply from demand in this decomposition: policy

rate cuts also stimulate loan demand, which could independently raise volumes and compress spreads. We therefore view this decomposition as informative rather than definitive, and focus the paper's core conclusions on the overall effect of cross-selling on pass-through.

5.7. Heterogeneity Across Bank Types and Policy Rate Environments

Since the cross-selling mechanism incentivizes lower deposit spreads at lower policy rates, a natural question is whether it has an effective lower bound. Figure 1 shows that the volume-weighted average spread between the policy rate and deposit rates does not wait for the policy rate to turn negative: it falls to essentially zero once the policy rate reaches 1.5 percentage points (2012–2014) and turns negative as the policy rate falls below 1 percentage point (2016–2018); Wang (2025) documents the same pattern for the United States, where deposit spreads likewise turn negative before policy rates do. We therefore classify the years with policy rates below 1.5 percentage points (2015–2018) as the low-rate period: these are the years in which the average bank no longer earns a positive margin on its deposits. The cut-off also allows for spreads being set against a somewhat longer-term benchmark, such as 6- or 12-month government bonds, whose yields lie somewhat above the policy rate. Two measurement choices matter for this classification. Our deposit rates are average effective rates on all household deposits, the rates relevant both for banks' funding costs and for transmission. Our benchmark is the policy rate: because Norwegian lending is almost entirely adjustable-rate, the replicating portfolio that Wang (2025) uses to net maturity-transformation profits out of the deposit spread differs little from the short rate here; benchmarking instead against zero-coupon government yields in the low-rate years, the volume-weighted average spread is -0.12 percentage points against the 12-month yield (against the policy rate: -0.08), 0.09 against the 3-year yield, and 0.37 and 0.89 against the 5- and 10-year yields, which embed a term premium that banks with adjustable-rate loan books do not earn.

Deposit spreads do move slightly into negative territory in these years: the mean bank-level spread is -0.12 percentage points, in line with the abundant evidence from negative-interest-rate policy episodes (e.g., Basten and Mariathasan, 2023; Eggertsson et al., 2024). What changes is the link between cross-selling potential and pass-through. In the full sample (Table 10, column 1), the triple interaction $PrB \times LowRate \times \Delta PR$ is -0.684 (SE $= 0.136$), far larger in absolute value than the positive interaction of 0.185 in the remaining years, so that the net slope in the low-rate years is -0.50: for the average bank the cross-selling mechanism does not merely weaken, it disappears; columns 2–5 show that the weakening is strongest for banks with fewer financial resources. Our explanation is that banks' capacity to sustain losses from their deposit business is limited: the mechanism has an effective lower bound, which binds sooner for banks with fewer resources. This is consistent with Wang (2025), who shows how the compression of deposit spreads at low rates reduces bank profits and net worth and can thereby constrain intermediation; in our setting, banks with fewer resources act a step earlier, by slowing the compression of their deposit spreads rather than by cutting loan supply.

Columns 2–5 of Table 10 test this by splitting the sample at the medians of respectively bank profitability (profits over total assets) and bank size (log total assets). The triple interaction $PrB \times LowRate \times \Delta PR$ captures how the cross-selling effect changes during low-rate periods. For less profitable banks, the triple interaction (-1.02) far exceeds the baseline interaction in absolute value, so the net slope turns negative when deposit spreads

turn negative (−0.80); for more profitable banks the offset (−0.42) is smaller, leaving a net slope of −0.23. The pattern is similar for size: for smaller banks the net slope turns negative (−0.90), while for larger banks the offset almost exactly cancels the baseline (net slope −0.03). The cross-sample differences point in the expected direction but are not statistically significant at conventional levels ($p = 0.12$ for the profitability split and $p = 0.10$ for the size split), so we read them as suggestive rather than conclusive.

These results suggest that resource constraints, rather than the disappearance of cross-selling incentives, explain the attenuation at low policy rates. Banks with greater capacity to absorb temporary deposit losses continue to act on their cross-selling incentives even when spreads are negative. One caveat is that profitability may itself be partly endogenous to cross-selling ability: banks that are better at converting depositors into borrowers may appear more profitable for that reason. The split-sample results should therefore be interpreted as suggestive of a resource-constraint mechanism rather than as cleanly identified causal effects.

5.8. Additional Robustness

Timing. Table A.2 adds the lead and the lag of the policy-rate change, each interacted with PrB, to the baseline specification. The contemporaneous interaction is unchanged (0.108–0.178), while the lead (0.028–0.073) and the lag (−0.033 to −0.042) are small and insignificant: the pass-through response is an impact response within the year of the rate change, with neither anticipation nor delayed catch-up. Because depositor demand and bank incentives respond to the same contemporaneous rate change, this exercise speaks to timing rather than to the sophistication concern of Section 5.3, which is why we report it here.

We examine the sensitivity of the pricing results to alternative specifications. Using levels of policy rates and deposit spreads rather than year-on-year changes yields effects of 1.9 to 2.4 basis points per SD (Table A.6), two to five times the baseline. Weighting observations by deposit volume produces estimates of 1.5 to 2.1 basis points per SD, and 6.3 basis points (significant at the 10% level) in the HHI-control specification (Table A.7), consistent with banks exercising greater market power in their larger local markets. Both checks confirm that our baseline first-differenced, unweighted specification is conservative.

A related concern is that policy rate changes may reflect macroeconomic conditions that independently affect deposit pricing. Table A.8 replaces observed rate changes with the narrative monetary policy surprises constructed by Holm et al. (2021). The point estimates are between roughly half and 85% of the corresponding baselines (e.g., 0.080 vs. 0.166 in the baseline specification) and less precise, as expected: with over 99% of deposits maturing within a year, the distinction between expected and surprise rate changes is largely moot for our setting (Drechsler et al., 2017).

5.9. External Validity

Our identification strategy relies on Norwegian administrative data, and few countries outside Scandinavia provide comparable relationship-level coverage. However, euro area survey evidence offers suggestive support. The European Central Bank (ECB)'s Single Supervisory Mechanism (SSM) in supervisory survey asks the 120 largest euro area banks whether cross-selling considerations affect their deposit and loan pricing. Banks that report considering cross-selling offer lower deposit spreads and charge higher loan spreads than those that

do not, consistent with loss-leader pricing, e.g., Figure 6. Because the survey data are available only for 2023, we cannot test whether these banks also exhibit higher deposit spread betas, but the cross-sectional pricing patterns align with our framework.[5] More broadly, the mechanism we document, banks internalizing future lending profits when setting deposit prices, should operate wherever households tend to consolidate banking relationships.

6. Conclusion

Banks' potential to cross-sell loans to their depositors dampens the pass-through of policy rate changes to deposit rate changes, thereby magnifying the deposits channel of monetary policy. Using administrative data covering the universe of Norwegian bank-household relationships, we show that within each bank-year, pass-through is lower for clients with greater cross-selling potential. These pricing differences amplify both deposit and loan growth responses to monetary policy. Our identification analysis — an Altonji–Oster analysis of selection on observables and unobservables together with a residualization exercise, and a decomposition showing that the relevant variation in cross-selling potential is specific to each bank-municipality combination rather than common to all banks in a municipality — indicates that the OLS estimates are conservative: omitted variable bias attenuates rather than inflates the estimated effects.

Cross-selling broadens the concept of market power central to the Deposits Channel framework. Where the existing literature measures deposit market power through local market concentration, we show that banks' ability to convert depositors into future borrowers constitutes a dynamic, relationship-level dimension of market power that independently reduces pass-through—reinforcing the deposits channel in changes even though cross-selling raises deposit rate levels. Both dimensions survive jointly in the same specification, indicating that they capture complementary sources of pricing power.

The mechanism is strongest away from the lower bound of policy rates: it weakens sharply during extended low-rate periods, and increasingly so for banks with fewer resources to absorb temporarily unprofitable deposit pricing, suggesting that balance-sheet capacity conditions banks' ability to act on their cross-selling incentives.

By linking the deposit and loan franchises, our results suggest that the transmission of monetary policy through banks depends not only on the structure of deposit markets but also on the composition of banks' depositor bases and the lending opportunities they represent. Understanding how these relationship-level incentives interact with other dimensions of bank market power, in other institutional settings and across the interest rate cycle, is a promising direction for future work.

[5]Specifically, banks were asked: "Is the pricing of this product affected by cross-selling considerations (deposits/funding)?" We combine the survey answers "partly" (rare) and "fully" to make the answer binary. Banks are split into two groups based on their answer to the question whether the pricing of new loans was affected by cross-selling considerations. Loan spreads are weighted averages across floating and fixed rate portfolios, using new business volumes as weights. Loan spreads are reported by banks as the average yield on the respective loan category less the respective funding costs, which should reflect the maturity. Deposit rates are computed based on net interest income flows and outstanding amounts. If a bank reported no impact of cross selling on floating (fixed rate) loans, but an impact for the other, all lending is considered impacted.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The bank–household relationship data, the household tax records, and the bank-level supervisory data used in this paper are confidential Norwegian administrative microdata that the authors are not permitted to share. The programs that reproduce all tables and figures, detailed instructions for their use, and a pseudo data set with the structure of the analysis data are available on Mendeley Data at [DOI TO BE INSERTED].

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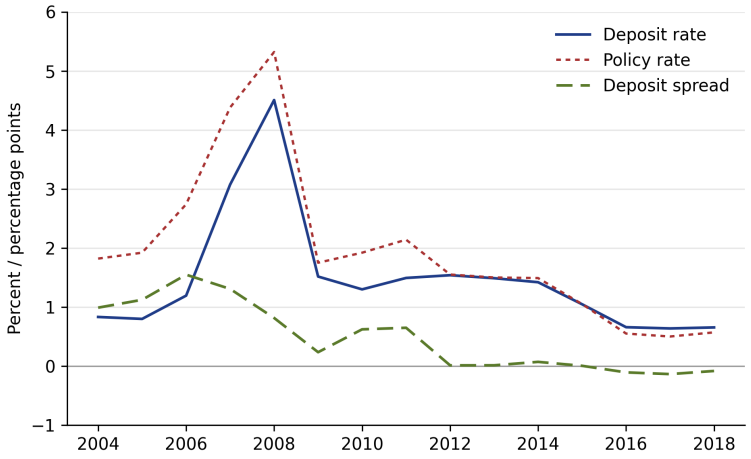


Fig. 1: Average Deposit Spreads in Norway over Time. Annual averages over the bank–municipality cells of the baseline sample, weighted by deposit volume, 2004–2018: the deposit rate and the policy rate in percent, and the deposit spread (policy rate minus deposit rate) in percentage points. The spread falls to zero once the policy rate reaches 1.5 percentage points and is negative from 2016 on.

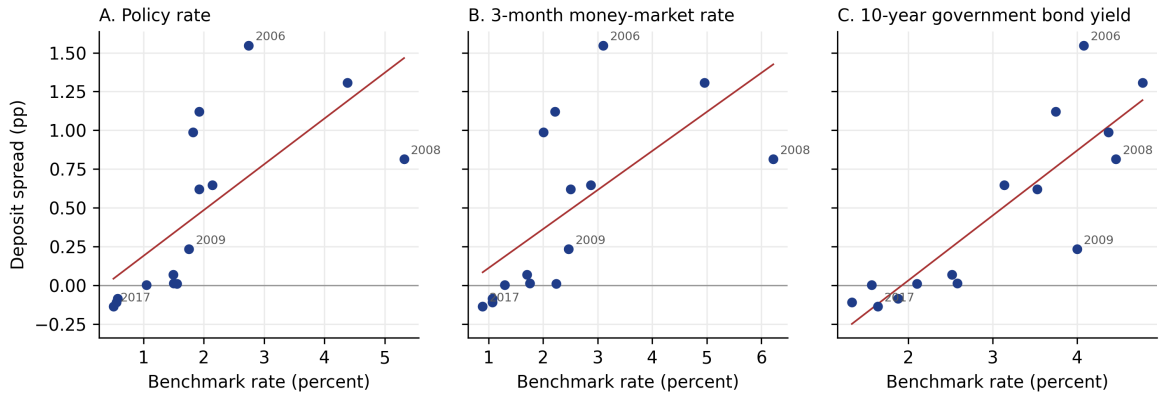


Fig. 2: Deposit Spread against Short- and Long-Term Benchmark Rates. Each marker is one year, 2004–2018. The vertical axis is the volume-weighted average deposit spread (policy rate minus deposit rate) of the baseline sample in percentage points, as in Figure 1; the horizontal axis is the annual average of the Norges Bank policy rate (panel A), the 3-month money-market rate (panel B), and the 10-year government bond yield (panel C), in percent. Lines are OLS fits. Sources: Norges Bank; OECD Main Economic Indicators.

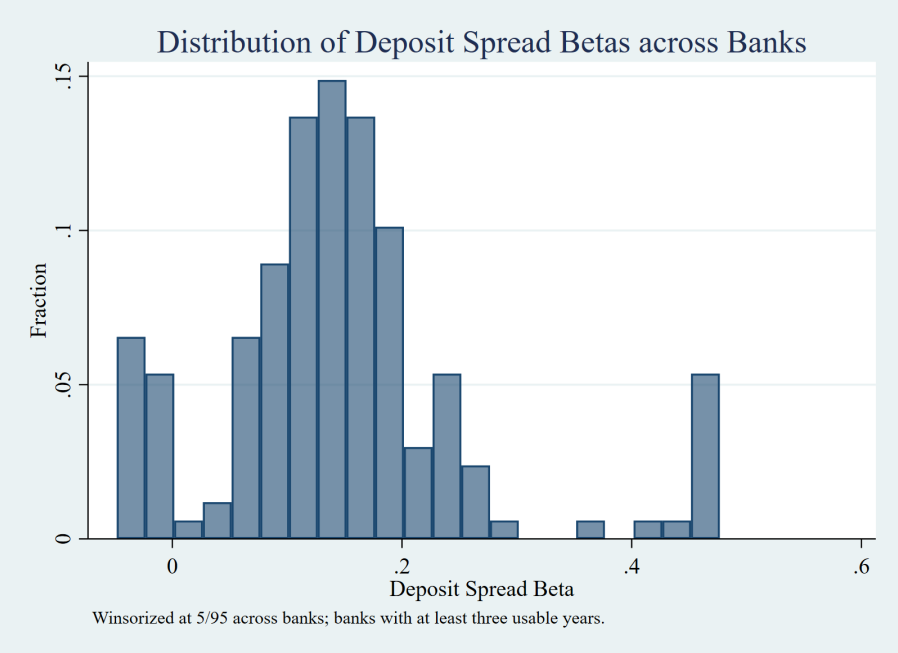


Fig. 3: Distribution of Pass-Through (Deposit Spread Betas) across Banks. Distribution of bank-level pass-through (deposit spread betas), estimated for each bank as the coefficient from regressing annual deposit-spread changes on policy-rate changes, 2004–2018, for the 168 banks with at least three usable observations; winsorized at the top and bottom 5% across banks.

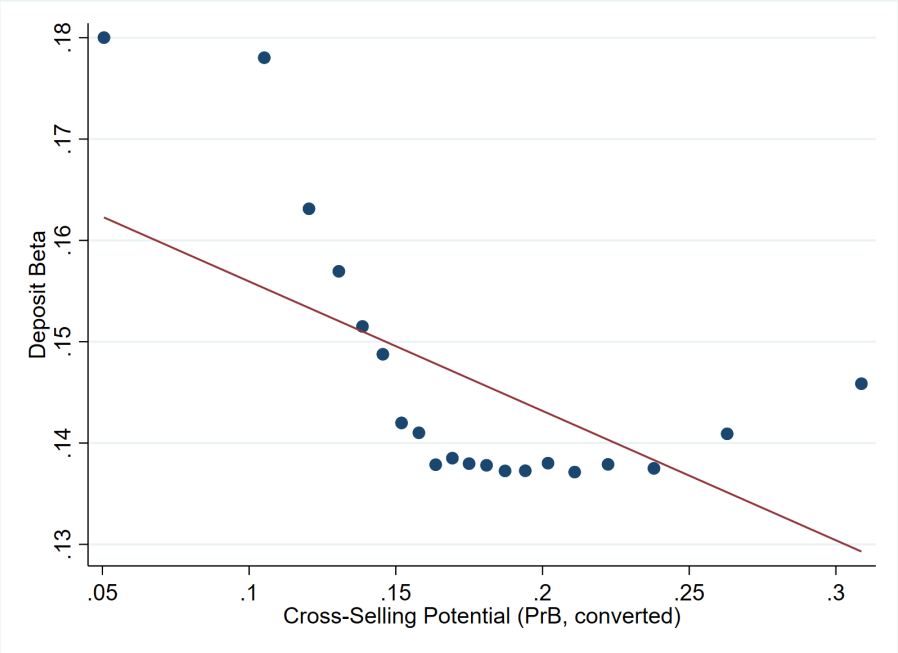


Fig. 4: Bank-Level Pass-Through (Deposit Betas) Plotted against Borrowing Propensity. Binned scatter of bank-level deposit betas against cross-selling potential (PrB), over bank–municipality–year observations grouped into 20 PrB bins, 2004–2018. Betas estimated per bank as in Figure 3 (at least three usable observations; winsorized at 5/95 across banks).

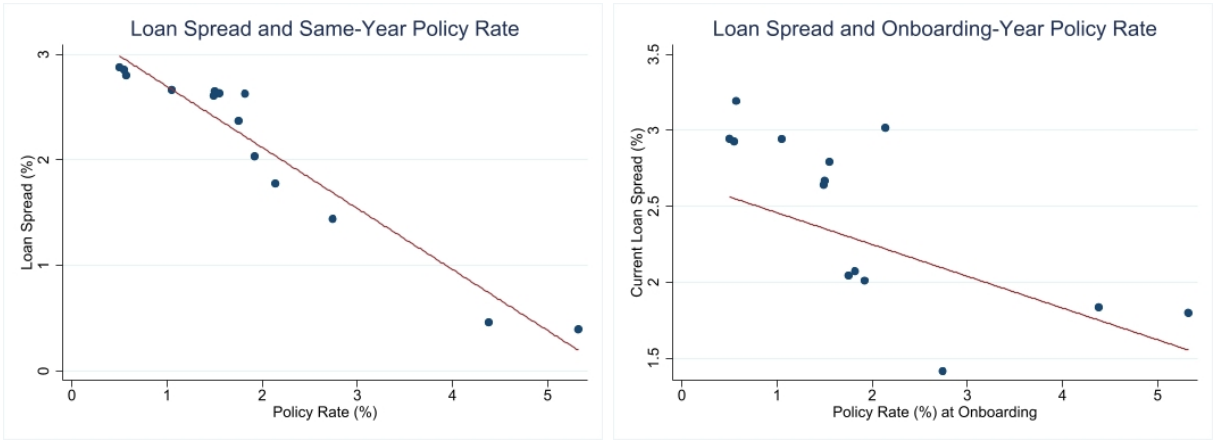


Fig. 5: Loan Spread by Same-Year and by Onboarding-Year Policy Rate. Average loan spread plotted against the policy rate in the year the loan is observed (left) and against the policy rate in the year the borrower was onboarded as a depositor (right).

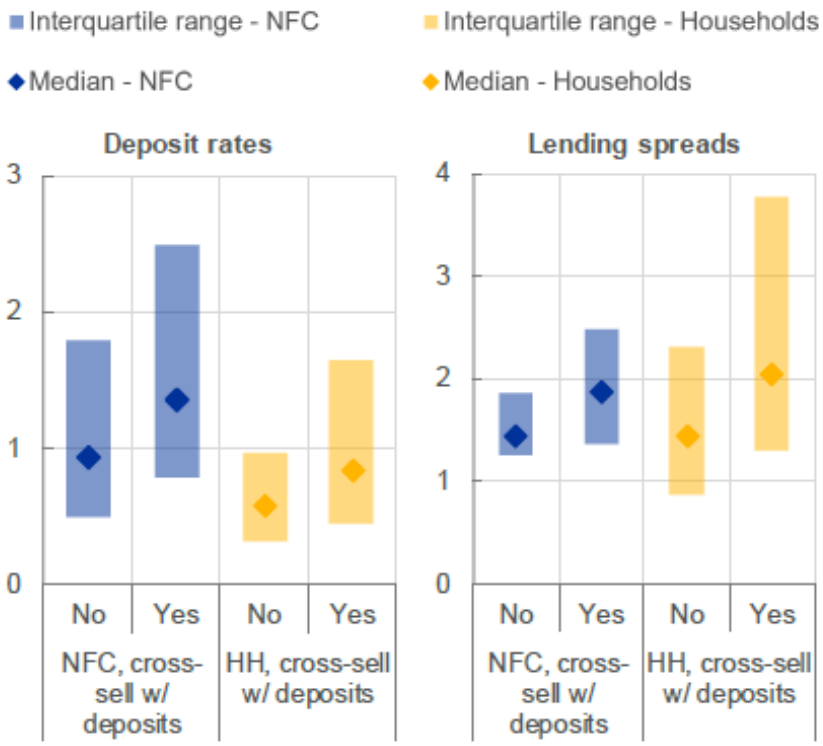


Fig. 6: Cross-Selling and Deposit Pricing in the Euro Area. Euro-area evidence on the relationship between cross-selling and deposit pricing, constructed from ECB data; see Section 5.9. The figure plots deposit *rates*; the higher deposit rates among banks that report cross-selling considerations correspond to lower deposit spreads.

Table 1: Summary Statistics

The table reports summary statistics for the main analysis samples, 2004–2018. Panel A: bank–household relationship level. Panel B: bank–municipality–year level, the estimation sample for the deposit-pricing analysis. Panels C1–C3: corporate loan-growth samples at the bank–municipality and bank–sector (NACE3, NACE5) levels; loan growth is in percent, LD denotes the log-difference and Sym the symmetric growth rate. Panel D: bank level; the deposit spread beta is the bank-specific coefficient from regressing annual deposit-spread changes on policy-rate changes. The deposit spread is the policy rate minus the deposit rate, in percentage points. Pr(borrow) = PrB is the conversion-based cross-selling measure: the estimated probability that an existing depositor subsequently takes a loan from the same bank (Section 3.2). Pr(mean debt), Pr(ln mean debt), and Pr(LS) are the corresponding estimated loan volume and loan-spread components used for the intensive-margin measures in Table 4. HHI is the deposit-market Herfindahl–Hirschman index at the municipality–year level; MS is the bank's municipal deposit market share. Deposit volumes in NOK. The Deposit Spread Beta row covers the 168 banks with at least three usable annual observations; other Panel D rows cover all 196 banks.

	Obs.	Mean	SD	Min	P50	Max
A. Relationship Level						
Policy Rate (PR)	43,654,911	1.931	1.378	0.500	1.550	5.320
Policy Rate Change (ΔPR)	43,654,911	−0.129	1.160	−3.570	−0.010	1.640
Deposit Spread (DS)	43,654,911	0.457	1.278	−2.435	0.442	2.856
Dep. Spread Change	43,654,911	−0.077	1.097	−5.292	0.000	5.292
Deposit Volume	43,654,911	135,410	302,281	0	20,194	2,001,353
Deposit Growth (log-diff)	30,154,294	−27.200	65.126	−100.000	−30.002	172.853
Pr(borrow) = PrB	43,654,911	0.180	0.072	−1.108	0.174	0.426
Pr(mean debt) in NOK	43,654,911	1,079,847	507,389	−1,351,384	1,113,093	15,100,000
Pr(ln mean debt)	43,654,911	13.222	0.773	9.844	13.304	33.573
Pr(LoanSpread LS)	43,654,911	4.989	1.991	−29.534	4.838	18.670
Pr(LS Overnight)	43,654,911	2.339	1.138	−12.005	2.163	10.001
Herfindahl–Hirschman Index HHI	43,654,911	0.291	0.121	0.114	0.264	0.864
Deposit Market Share MS	39,551,680	0.212	0.287	0.000	0.052	0.998
B. Bank–Municipality Level						
Policy rate (%)	363,204	1.885	1.281	0.50	1.55	5.32
ΔPol.Rate (pp)	363,204	−0.100	1.080	−3.57	−0.01	1.64
Dep.Spread (pp)	363,204	0.480	0.926	−2.860	0.452	2.856
ΔDep.Spread (pp)	363,204	−0.082	0.695	−5.567	−0.022	5.532
Deposit Volume (NOK)	363,204	2.20×10^{7}	5.44×10^{8}	0	32,406	1.25×10^{11}
PrB	363,204	0.178	0.050	−0.924	0.174	0.418
HHI deposits	363,204	0.381	0.132	0.114	0.361	0.864
Market Share	353,350	0.016	0.085	0	0.0001	0.998
Number of clients	363,204	192	2,049	1	7	277,994

continues

Table 1 (continued)

	Obs.	Mean	SD	Min	P50	Max
C1. Loan Growth at Bank–Municipality Level						
Loan Growth (LD)	91,293	−3.90	93.92	−407.92	−3.67	389.38
Loan Growth (Sym)	91,293	−3.89	64.36	−193.37	−3.67	192.11
Ln(New Loan Volume)	23,439	12.98	3.79	−0.69	13.41	25.00
Bank-Level Pr(borrow)	130,652	0.19	0.03	0.04	0.20	0.27
Bank-Level HHI	130,652	0.31	0.08	0.16	0.29	0.70
C2. Loan Growth at Bank–Sector (NACE3) Level						
Loan Growth (LD)	125,923	−3.06	91.75	−414.90	−3.08	368.25
Loan Growth (Sym)	125,923	−2.63	63.49	−193.84	−3.08	190.21
Ln(New Loan Volume)	26,622	12.69	3.59	−0.69	13.15	25.05
Bank-Level Pr(borrow)	196,035	0.20	0.03	0.04	0.20	0.27
Bank-Level HHI	196,035	0.33	0.09	0.16	0.30	0.70
C3. Loan Growth at Bank–Sector (NACE5) Level						
Loan Growth (LD)	188,366	−6.54	97.31	−457.54	−4.60	381.74
Loan Growth (Sym)	188,366	−5.25	65.26	−195.94	−4.60	191.42
Ln(New Loan Volume)	46,327	12.41	3.58	−0.69	12.93	25.05
Bank-Level Pr(borrow)	346,292	0.20	0.03	0.04	0.20	0.27
Bank-Level HHI	346,292	0.32	0.09	0.16	0.30	0.70
D. Bank Level						
Deposit Spread Beta	168	0.147	0.113	−0.048	0.140	0.469
Pr(borrow)	196	0.15	0.05	−0.02	0.15	0.29
HHI	196	0.26	0.05	0.18	0.25	0.60

Table 2: Expected Borrowing Propensity, Amount and Spread of Each Deposit Relationship

Linear probability model on the deposit-conditional sample (existing depositors). Column 3 is conditional on holding a loan. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) I(borrow)	(2) Ln(MeanDebt)	(3) Loan Spread
I(Parent)	-0.051***	0.603***	-0.210***
	(0.000)	(0.002)	(0.003)
I(Retired)	0.026***	-2.536***	0.660***
	(0.000)	(0.001)	(0.002)
I(age<30)	0.145***	1.066***	-0.403***
	(0.000)	(0.001)	(0.002)
Ln(Inc)	0.019***	0.997***	-0.135***
	(0.000)	(0.001)	(0.002)
Ln(Wealth)	0.001***	0.116***	-0.279***
	(0.000)	(0.000)	(0.001)
Ln(Deposits)	0.001***	-0.004***	-0.000
	(0.000)	(0.000)	(0.000)
No. of banks of HH	-0.014***	0.508***	-0.161***
	(0.000)	(0.000)	(0.001)
HHI (Deposit)	0.167***	-2.258***	0.948***
	(0.000)	(0.006)	(0.008)
Constant	-0.143***	-3.398***	7.654***
	(0.001)	(0.010)	(0.021)
Observations	53,594,650	53,594,650	13,056,325
R^2	0.040	0.228	0.056

Table 3: Borrowing Propensity and the Deposit Spread Beta

Outcome is the year-on-year change in the deposit spread. PrB is the conversion-based cross-selling measure. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Δ Dep. Spread	(2) Δ Dep. Spread	(3) Δ Dep. Spread	(4) Δ Dep. Spread
PrB × ΔPol.Rate	0.117	0.166***	0.102***	0.177***
	(0.073)	(0.032)	(0.037)	(0.032)
PrB	-0.185***	-0.333***	-0.394***	-0.342***
	(0.069)	(0.027)	(0.031)	(0.028)
Δ Pol. Rate	0.109***			
	(0.019)			
HHI × ΔPol.Rate			0.051***	
			(0.012)	
HHI			0.052***	
			(0.010)	
MS × ΔPol.Rate				0.001
				(0.011)
MS				0.025***
				(0.007)
Constant	-0.036**	-0.020***	-0.028***	-0.019***
	(0.016)	(0.005)	(0.005)	(0.005)
Observations	363,212	363,204	363,204	353,350
R^2	0.041	0.265	0.265	0.264
Bank×Year FE	No	Yes	Yes	Yes
Effect of 1SD of PrB	0.006	0.008***	0.005***	0.009***
Effect of 1SD of HHI			0.007***	
Effect of 1SD of MS				0.000

Table 4: Volume- and Spread-Adjusted Cross-Selling Potential and the Deposit Spread Beta

CSP_{vol}: estimated borrowing propensity times estimated log loan volume (columns 1–4); $CSP_{vol \times LS}$ additionally times estimated loan spread (columns 5–8). Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	PrB×Ln(V)				PrB×Ln(V)×LS			
	(1) ΔDS	(2) ΔDS	(3) ΔDS	(4) ΔDS	(5) ΔDS	(6) ΔDS	(7) ΔDS	(8) ΔDS
CSP_{vol} × ΔPol.Rate	0.009*	0.011***	0.008***	0.012***				
	(0.005)	(0.002)	(0.003)	(0.002)				
CSP_{vol}	-0.024***	-0.029***	-0.030***	-0.030***				
	(0.004)	(0.002)	(0.002)	(0.002)				
$CSP_{vol \times LS}$ × ΔPol.Rate					-0.000	0.003***	0.002***	0.003***
					(0.001)	(0.000)	(0.000)	(0.000)
$CSP_{vol \times LS}$					0.000	-0.002***	-0.002***	-0.002***
					(0.001)	(0.000)	(0.000)	(0.000)
Δ Pol. Rate	0.110***				0.131***			
	(0.016)				(0.017)			
HHI × ΔPol.Rate			0.056***				0.058***	
			(0.011)				(0.010)	
HHI			0.031***				-0.005	
			(0.009)				(0.009)	
MS × ΔPol.Rate				0.007				0.009
				(0.011)				(0.011)
MS				0.017**				0.006
				(0.007)				(0.007)
Constant	-0.017	-0.015***	-0.023***	-0.014***	-0.073***	-0.061***	-0.059***	-0.060***
	(0.014)	(0.005)	(0.005)	(0.005)	(0.013)	(0.004)	(0.005)	(0.004)
Observations	363,212	363,204	363,204	353,350	363,212	363,204	363,204	353,350
R^2	0.041	0.265	0.265	0.264	0.041	0.264	0.264	0.263
Bank×Year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Effect of 1SD of CSP	0.006*	0.007***	0.005***	0.008***	-0.001	0.015***	0.014***	0.016***
Effect of 1SD of HHI			0.007***				0.008***	
Effect of 1SD of MS				0.001				0.001

Table 5: Own-Bank versus Other-Banks Cross-Selling Potential

The dependent variable is the change in the deposit spread. Other-banks PrB is the leave-one-out average cross-selling potential of all other banks in the same municipality-year, unweighted in columns (2)–(3) and deposit-volume-weighted in column (4); column (5) adds deposit market concentration (HHI × ΔPolRate). Level terms of each PrB measure and of the HHI are included; the policy-rate change is absorbed by the bank-year fixed effects. Standard errors clustered at the bank-year level in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

	(1)	(2)	(3)	(4)	(5)
Own PrB × ΔPolRate	0.166***		0.121***	0.123***	0.102***
	(0.032)		(0.036)	(0.036)	(0.037)
Other-banks PrB (leave-one-out) × ΔPolRate		0.313***	0.185***		0.005
		(0.058)	(0.063)		(0.118)
Other-banks PrB (LOO, deposit-weighted) × ΔPolRate				0.168***	
				(0.050)	
Deposit HHI × ΔPolRate					0.050**
					(0.022)
Bank–Year FE	Yes	Yes	Yes	Yes	Yes
Observations	363,204	363,204	363,204	363,204	363,204
R-squared	0.265	0.264	0.265	0.265	0.265

Table 6: Cross-Selling Potential and the Deposit Growth Beta

Deposit Growth (DG) is the year-on-year log difference in deposit volume; columns 1–4 raw, columns 5–8 winsorized at the 10th/90th percentiles. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	DG, raw				DG, 10% winsorized			
	(1) DG	(2) DG	(3) DG	(4) DG	(5) DG	(6) DG	(7) DG	(8) DG
PrB × ΔPol.Rate	-13.725	-10.947	-27.118**	-5.848	-3.872	-2.772	-14.131*	0.174
	(10.087)	(9.477)	(11.539)	(8.817)	(6.681)	(6.633)	(8.073)	(6.252)
PrB	295.952***	299.189***	386.984***	227.656***	199.652***	203.759***	265.124***	156.576***
	(12.545)	(11.970)	(13.564)	(11.492)	(8.427)	(8.265)	(9.382)	(8.004)
Δ Pol. Rate	3.615*				1.361			
	(2.155)				(1.374)			
HHI × ΔPol.Rate			10.066**				7.080**	
			(4.242)				(2.952)	
HHI			-75.447***				-52.732***	
			(5.194)				(3.520)	
MS × ΔPol.Rate				-10.930				-6.424
				(8.453)				(5.558)
MS				350.356***				233.243***
				(12.177)				(7.452)
Constant	-47.296***	-48.198***	-35.529***	-41.606***	-29.066***	-29.934***	-21.079***	-25.644***
	(2.576)	(2.190)	(2.463)	(2.112)	(1.690)	(1.511)	(1.681)	(1.466)
Observations	282,625	282,620	282,620	275,077	282,625	282,620	282,620	275,077
R^2	0.002	0.012	0.013	0.021	0.002	0.012	0.013	0.021
Bank×Year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Effect of 1SD of PrB	-0.685	-0.546	-1.352**	-0.293	-0.193	-0.138	-0.705*	0.009
Effect of 1SD of HHI			1.332**				0.937**	
Effect of 1SD of MS				-1.023				-0.601

Table 7: Corporate Loan Growth at the Bank–Municipality Level

Columns 1–3: NFC loan growth (log difference); 4–6: symmetric growth; 7–9: log new loan volume. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	LG (LD)			LG (Sym)			Ln(new loans)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PrB × ΔPol.Rate	-55.363***		-66.257***	-39.366***		-47.313***	-2.894**		-3.249**
	(13.987)		(24.629)	(9.484)		(16.568)	(1.168)		(1.638)
PrB	73.577***		132.175***	64.589***		112.295***	7.199***		11.550***
	(16.808)		(23.913)	(11.346)		(16.302)	(1.866)		(2.929)
HHI × ΔPol.Rate		-13.590***	5.621		-9.614***	4.140		-0.529	0.028
		(4.508)	(8.171)		(3.081)	(5.532)		(0.385)	(0.575)
HHI		4.761	-29.126***		5.105	-23.699***		1.127*	-2.145**
		(5.399)	(7.660)		(3.707)	(5.329)		(0.679)	(1.085)
Constant	-15.220***	-1.858	-17.225***	-13.256***	-1.956*	-14.888***	11.213***	12.557***	11.017***
	(3.135)	(1.670)	(3.221)	(2.113)	(1.146)	(2.169)	(0.350)	(0.197)	(0.351)
Observations	57,403	57,403	57,403	57,403	57,403	57,403	11,360	11,360	11,360
R^2	0.117	0.117	0.117	0.114	0.113	0.115	0.342	0.339	0.342
Muni×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Effect of 1SD of PrB	-1.559***		-1.865***	-1.108***		-1.332***	-0.085**		-0.096**
Effect of 1SD of HHI		-1.064***	0.440		-0.753***	0.324		-0.043	0.002

Table 8: Corporate Loan Growth at the Bank–Sector Level (NACE3)

Columns 1–3: NFC loan growth (log difference); 4–6: symmetric growth; 7–9: log new loan volume. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	LG (LD)			LG (Sym)			Ln(new loans)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PrB × ΔPol.Rate	-21.162**		-63.001***	-13.229**		-44.325***	1.342**		-4.893***
	(9.940)		(20.214)	(6.366)		(13.269)	(0.680)		(1.172)
PrB	38.854***		130.560***	27.034***		99.516***	7.841***		9.548***
	(11.125)		(21.712)	(7.394)		(14.400)	(1.335)		(2.325)
HHI × ΔPol.Rate		-1.540	14.987***		-0.479	11.160***		1.133***	2.242***
		(2.615)	(5.335)		(1.747)	(3.661)		(0.203)	(0.364)
HHI		-0.282	-33.180***		-1.138	-26.210***		2.143***	-0.433
		(3.012)	(5.950)		(2.063)	(4.051)		(0.423)	(0.738)
Constant	-9.385***	-1.169	-16.923***	-6.494***	-0.470	-12.442***	11.478***	12.445***	10.684***
	(2.223)	(1.011)	(2.787)	(1.474)	(0.690)	(1.831)	(0.207)	(0.102)	(0.241)
Observations	103,216	103,216	103,216	103,216	103,216	103,216	18,500	18,500	18,500
R^2	0.061	0.061	0.062	0.064	0.063	0.064	0.342	0.342	0.345
NACE3×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Effect of 1SD of PrB	-0.589**		-1.754***	-0.368**		-1.234***	0.039**		-0.143***
Effect of 1SD of HHI		-0.142	1.386***		-0.044	1.032***		0.104***	0.206***

Table 9: How Components of Cross-Selling Flow Profits Vary with the Policy Rate

Outcomes: propensity of a deposit relationship to later become a loan relationship; log mean loan volume; loan spread. Columns 1–3 baseline, 4–6 with bank FE, 7–9 controlling for later NPL status. Sample sizes differ by outcome. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	Baseline			Bank FE			NPL control		
	(1) I(loan)	(2) Ln(Vol)	(3) LS	(4) I(loan)	(5) Ln(Vol)	(6) LS	(7) I(loan)	(8) Ln(Vol)	(9) LS
Panel A. Flow Profits by Policy-Rate Change									
Δ Pol. Rate	-0.029***	-0.059***	-0.264***	-0.046***	-0.137***	-0.278***	-0.016***	0.007***	-0.261***
	(0.000)	(0.001)	(0.002)	(0.000)	(0.001)	(0.002)	(0.000)	(0.001)	(0.002)
I(later NPL)							0.547***	3.066***	0.155***
							(0.000)	(0.002)	(0.010)
Constant	0.548***	11.297***	2.080***	0.536***	11.234***	2.068***	0.439***	10.183***	2.068***
	(0.000)	(0.001)	(0.003)	(0.000)	(0.001)	(0.003)	(0.000)	(0.002)	(0.003)
Observations	14,519,483	8,252,937	3,061,556	14,519,483	8,252,936	3,061,552	14,519,483	8,252,937	3,061,556
R^2	0.006	0.001	0.007	0.162	0.169	0.031	0.212	0.201	0.007
Panel B. Weighted by Mean Deposit Volume									
Δ Pol. Rate	-0.044***	-0.135***	-0.225***	-0.055***	-0.155***	-0.217***	-0.026***	-0.065***	-0.221***
	(0.000)	(0.002)	(0.003)	(0.000)	(0.002)	(0.004)	(0.000)	(0.002)	(0.003)
I(later NPL)							0.579***	3.179***	0.139***
							(0.000)	(0.004)	(0.017)
Constant	0.504***	11.431***	2.174***	0.496***	11.416***	2.180***	0.398***	10.275***	2.165***
	(0.000)	(0.003)	(0.007)	(0.000)	(0.003)	(0.007)	(0.000)	(0.003)	(0.007)
Observations	14,519,483	8,252,937	3,061,556	14,519,483	8,252,936	3,061,552	14,519,483	8,252,937	3,061,556
R^2	0.013	0.003	0.005	0.121	0.208	0.030	0.228	0.225	0.005

Table 10: Heterogeneity in the Cross-Selling Effect by Bank Financial Strength during Low-Rate Periods
LowRate indicates the years with policy rates below 1.5 percentage points (2015–2018; see Section 5.7). Column (1) reports the pooled full-sample specification; columns (2)–(5) split the sample at the median of bank profitability and size. The t/p rows test whether the triple interaction differs across subsamples; the differences are not significant at the 5% level ($p = 0.12$ profit split, $p = 0.10$ size split). Two-way clustering on bank and year. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) All banks	(2) Low Prof/TA	(3) High Prof/TA	(4) Small	(5) Large
PrB	-0.394***	-0.404***	-0.400***	-0.407***	-0.398***
	(0.061)	(0.084)	(0.043)	(0.048)	(0.083)
PrB × ΔPR	0.185***	0.214***	0.182***	0.136***	0.258***
	(0.043)	(0.048)	(0.040)	(0.033)	(0.059)
LowRate × PrB	0.080	0.027	0.123	-0.013	0.195
	(0.083)	(0.098)	(0.127)	(0.065)	(0.113)
PrB × LowRate × ΔPR	-0.684***	-1.016***	-0.416	-1.037***	-0.292
	(0.136)	(0.123)	(0.256)	(0.180)	(0.299)
t-stat (difference)	–	1.648	1.648	1.793	1.793
p-value (difference)	–	0.123	0.123	0.096	0.096
Observations	363,204	154,710	195,133	146,650	203,193
R^2	0.265	0.279	0.247	0.259	0.264
Bank×Year FE	Yes	Yes	Yes	Yes	Yes

Appendix A. Additional figures and tables

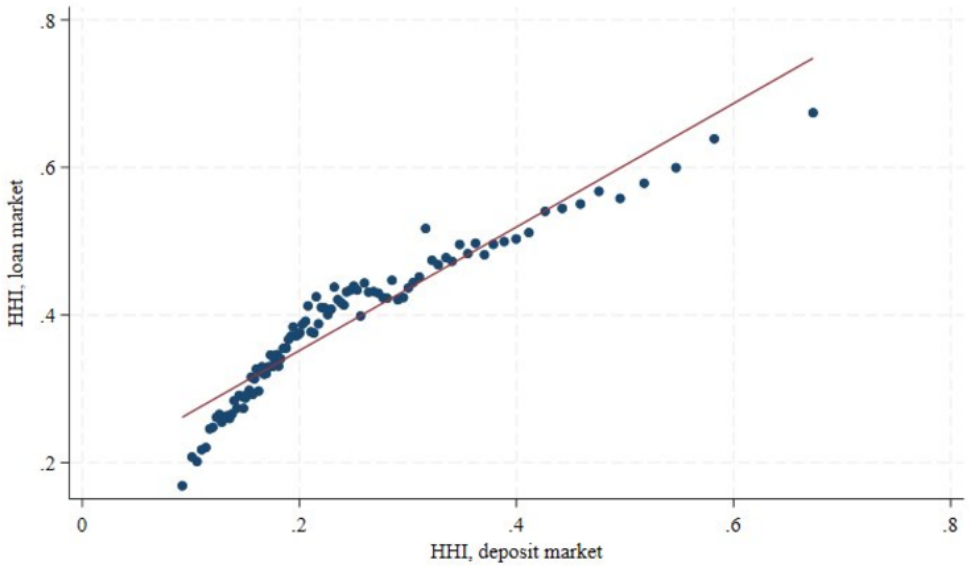


Fig. A.1: HHI in Loan vs. Deposit Markets. Municipality-level loan-market HHI plotted against deposit-market HHI; the correlation is approximately 0.90.

Table A.1: Altonji–Oster Comparison and Residualized Cross-Selling Potential

Columns (1)–(2): Altonji–Oster comparison; the long specification adds the four sophistication proxies (education, residential mobility, stock-market participation, and the share of wealth held in stocks and bonds), each in levels and interacted with the policy rate change. Columns (3)–(4): PrB is orthogonalized within bank-year against age, income, and wealth (column 3) and additionally against the sophistication proxies (column 4); the residual replaces PrB. Level terms are included throughout but suppressed for space. Bank-year fixed effects and the control set jointly absorb 51% of the variance of PrB. Standard errors clustered at the bank-year level in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

	(1) Short	(2) Long	(3) Residualized	(4) Residualized
PrB × ΔPolRate	0.166***	0.215***		
	(0.032)	(0.036)		
Educ. × ΔPolRate		0.016***		
		(0.003)		
Move × ΔPolRate		−0.102***		
		(0.011)		
Stock part. × ΔPolRate		−0.043***		
		(0.006)		
Stock/bond share × ΔPolRate		0.006		
		(0.009)		
Resid. PrB × ΔPolRate			0.223***	0.217***
			(0.041)	(0.042)
Residualized against	–	–	Age, Inc., Wealth	Full
Bank–Year FE	Yes	Yes	Yes	Yes
Observations	363,204	359,961	359,828	359,650
R-squared	0.2648	0.2664	0.2641	0.2642

Table A.2: Future-Policy Placebo: Leads and Lags of the Policy-Rate Change

Sample restricted to years for which both the lead and the lag of the policy-rate change exist. The contemporaneous interaction is significant; leads and lags are null, confirming that the pass-through response is concentrated in the year of the rate change. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Δ Dep. Spread	(2) Δ Dep. Spread	(3) Δ Dep. Spread	(4) Δ Dep. Spread
PrB × ΔPol.Rate	0.124*	0.166***	0.108***	0.178***
	(0.072)	(0.034)	(0.039)	(0.034)
PrB × ΔPol.Rate$_{t+1}$	0.073	0.031	0.035	0.028
	(0.054)	(0.031)	(0.031)	(0.031)
PrB × ΔPol.Rate$_{t-1}$	-0.042	-0.033	-0.033	-0.034
	(0.061)	(0.027)	(0.027)	(0.028)
PrB	-0.209***	-0.206***	-0.280***	-0.214***
	(0.069)	(0.032)	(0.037)	(0.033)
ΔPol.Rate	0.097***			
	(0.018)			
ΔPol.Rate$_{t+1}$	0.074***			
	(0.014)			
ΔPol.Rate$_{t-1}$	-0.126***			
	(0.016)			
HHI × ΔPol.Rate			0.044***	
			(0.013)	
HHI			0.059***	
			(0.011)	
MS × ΔPol.Rate				0.005
				(0.011)
MS				0.018**
				(0.008)
Constant	-0.051***	-0.057***	-0.065***	-0.055***
	(0.016)	(0.006)	(0.006)	(0.006)
Observations	285,260	285,251	285,251	277,433
R^2	0.141	0.275	0.275	0.274
Bank×Year FE	No	Yes	Yes	Yes

Table A.3: Loan-Market Concentration and the Deposit Spread Beta

Loan HHI is the Herfindahl–Hirschman index of loan-market shares at the municipality–year level. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Δ Dep. Spread	(2) Δ Dep. Spread	(3) Δ Dep. Spread	(4) Δ Dep. Spread
PrB × ΔPol.Rate	0.137	0.103***	0.102***	0.112***
	(0.092)	(0.035)	(0.037)	(0.035)
PrB	-0.313***	-0.360***	-0.394***	-0.370***
	(0.080)	(0.030)	(0.031)	(0.030)
ΔPol.Rate	0.111***			
	(0.017)			
Loan HHI × ΔPol.Rate	-0.029	0.068***	0.065***	0.072***
	(0.034)	(0.013)	(0.023)	(0.014)
Loan HHI	0.171***	0.034***	-0.031	0.037***
	(0.042)	(0.013)	(0.019)	(0.013)
HHI × ΔPol.Rate			0.006	
			(0.020)	
HHI			0.068***	
			(0.014)	
MS × ΔPol.Rate				0.001
				(0.011)
MS				0.024***
				(0.007)
Constant	-0.053***	-0.024***	-0.028***	-0.022***
	(0.014)	(0.005)	(0.005)	(0.005)
Observations	363,212	363,204	363,204	353,350
R^2	0.042	0.265	0.265	0.264
Bank×Year FE	No	Yes	Yes	Yes
Effect of 1SD of PrB	0.007	0.005***	0.005***	0.006***

Table A.4: Corporate Loan Growth at the Bank–Sector Level (NACE5)

Columns 1–3: NFC loan growth (log difference); 4–6: symmetric growth; 7–9: log new loan volume. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	LG (LD)			LG (Sym)			Ln(new loans)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PrB × ΔPol.Rate	-18.323**		-41.827**	-13.938**		-30.371**	1.765***		-3.041***
	(9.161)		(18.112)	(5.934)		(11.841)	(0.514)		(0.902)
PrB	40.842***		112.894***	26.174***		84.741***	9.363***		13.056***
	(9.750)		(18.088)	(6.409)		(11.822)	(1.042)		(1.783)
HHI × ΔPol.Rate		-2.170	8.865*		-1.769	6.249*		1.079***	1.725***
		(2.473)	(4.924)		(1.636)	(3.289)		(0.157)	(0.283)
HHI		1.453	-27.069***		-0.571	-21.979***		2.194***	-1.320**
		(2.720)	(5.092)		(1.831)	(3.398)		(0.338)	(0.581)
Constant	-12.954***	-5.045***	-18.554***	-8.691***	-3.102***	-13.233***	11.013***	12.141***	10.255***
	(1.939)	(0.895)	(2.347)	(1.272)	(0.601)	(1.526)	(0.159)	(0.080)	(0.185)
Observations	155,637	155,637	155,637	155,637	155,637	155,637	31,713	31,713	31,713
R^2	0.086	0.086	0.087	0.086	0.086	0.086	0.365	0.363	0.367
NACE5×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Effect of 1SD of PrB	-0.493**		-1.125**	-0.375**		-0.817**	0.050***		-0.086***
Effect of 1SD of HHI		-0.190	0.775*		-0.155	0.546*		0.094***	0.151***

Table A.5: Loan-Pricing Implications of a Loan Later Becoming Non-Performing

Dependent variable: loan spread. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) LS	(2) LS	(3) LS	(4) LS
I(later NPL)	0.478***		0.246***	
	(0.022)		(0.022)	
I(later NPL) v2		0.350***		0.224***
		(0.026)		(0.026)
Constant	2.120***	2.151***	2.149***	2.162***
	(0.008)	(0.008)	(0.008)	(0.008)
Observations	259,819	259,819	259,818	259,818
R^2	0.002	0.001	0.050	0.049

Table A.6: Policy-Rate Levels instead of Changes

Outcome is the deposit-spread level; PR is the policy-rate level. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) DS	(2) DS	(3) DS	(4) DS
PrB × PR	-0.228**	0.347***	0.445***	0.347***
	(0.089)	(0.027)	(0.031)	(0.027)
PrB	1.356***	-1.489***	-2.020***	-1.484***
	(0.243)	(0.061)	(0.071)	(0.062)
PR	0.356***			
	(0.023)			
HHI × PR			-0.081***	
			(0.009)	
HHI			0.463***	
			(0.020)	
MS × PR				-0.021**
				(0.009)
MS				-0.139***
				(0.021)
Constant	-0.306***	0.676***	0.619***	0.669***
	(0.057)	(0.006)	(0.006)	(0.006)
Observations	436,523	436,502	436,502	425,089
R^2	0.181	0.548	0.550	0.547
Bank×Year FE	No	Yes	Yes	Yes
Effect of 1SD of PrB	-0.012**	0.019***	0.024***	0.019***

Table A.7: Deposit-Volume-Weighted Regressions

Regressions weighted by deposit volume. Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Δ Dep. Spread	(2) Δ Dep. Spread	(3) Δ Dep. Spread	(4) Δ Dep. Spread
PrB × ΔPol.Rate	0.165	0.415***	1.280*	0.305***
	(0.338)	(0.135)	(0.682)	(0.064)
PrB	-0.291	0.135	0.270	0.111
	(0.280)	(0.119)	(0.522)	(0.091)
ΔPol.Rate	0.091			
	(0.074)			
HHI × ΔPol.Rate			-0.218	
			(0.144)	
HHI			-0.035	
			(0.107)	
MS × ΔPol.Rate				0.026
				(0.019)
MS				0.007
				(0.011)
Constant	0.003	-0.088***	-0.084	-0.086***
	(0.052)	(0.021)	(0.062)	(0.019)
Observations	310,303	310,295	310,295	301,845
R^2	0.162	0.868	0.869	0.864
Bank×Year FE	No	Yes	Yes	Yes
Effect of 1SD of PrB	0.008	0.021***	0.063*	0.015***

Table A.8: Monetary-Policy Surprises and the Deposit Spread Beta

The policy-rate change is replaced with the narrative monetary-policy surprise measure of Holm et al. (2021). Standard errors clustered by bank×year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	(1) DS	(2) DS	(3) DS	(4) DS
PrB × MP Shock	0.343*	0.080	0.088	0.079
	(0.182)	(0.069)	(0.076)	(0.070)
PrB	-0.198***	-0.353***	-0.411***	-0.362***
	(0.073)	(0.028)	(0.032)	(0.028)
MP Shock	0.121***			
	(0.046)			
HHI × MP Shock			-0.006	
			(0.024)	
HHI			0.048***	
			(0.010)	
MS × MP Shock				-0.043**
				(0.019)
MS				0.031***
				(0.007)
Constant	-0.071***	-0.021***	-0.029***	-0.020***
	(0.016)	(0.005)	(0.005)	(0.005)
Observations	363,212	363,204	363,204	353,350
R^2	0.012	0.265	0.265	0.264
Bank×Year FE	No	Yes	Yes	Yes